

The Effects of Request Alerts on the Diversity and Visibility of Community Notes

Keywords: Crowdsourced Fact-Checking, Request Alerts, Interface Cues, Contributor Diversity, Note Visibility

Extended Abstract

Several major social media platforms have shifted toward crowd-sourced fact-checking systems like Community Notes to combat misinformation at scale (Wojcik et al., 2022; Allen et al., 2021). However, these systems face two persistent criticisms: content is not equally vetted, with contributor biases shaping which posts receive scrutiny (Allen et al., 2022; Renault et al., 2025), and only a small fraction of submitted notes achieve sufficient consensus to be displayed to the public (Truong et al., 2025). To address these challenges, X allows users to request community notes for specific posts. When sufficient requests accumulate, X displays a request alert visible only to top writers for a 24-hour window. Drawing on the interface cue literature (Kim and Sundar, 2011), we conceptualize request alerts as a hybrid cue combining attention signaling with social signaling.

Data and Methods. Using publicly available Community Notes data, we reconstructed the historical publication status of 1.96 million notes, the dynamic top writer status of contributors, and the timing of 6.3 million requests. We inferred request alert presence at the time each note was written by jointly modeling three conditions: the writer held top writer status (Currently Rated Helpful [CRH] ratio $\geq 4\%$ and writing impact score ≥ 10), the target post had accumulated at least five requests, and the note was submitted within 24 hours of the fifth request. This improves upon Chuai et al. (2025), who characterize notes that emerge following requests but rely on overall CRH ratios; our study additionally accounts for dynamic writing impact scores and examines how alerts shape upstream contributor behavior.

After restricting to English-language notes and requiring sufficient repeated measures, our dataset comprises 318 top writers contributing 54,874 notes (10,432 request-alerted; 44,442 non-alerted) on 51,089 X posts. We applied topic modeling using Twitter4SSE embeddings, UMAP, and HDBSCAN, yielding 239 fine-grained clusters that two authors independently coded into seven broad categories (Table 1). We defined topic shift Φ_i as the cosine distance between note i 's embedding and the mean embedding of the writer's five preceding notes, capturing how far each note deviates from the writer's recent topical interests.

Findings. Our analysis reveals a nuanced relation between individual diversification and aggregate concentration. At the individual level, request alerts consistently redirected writers toward content outside their organic interests: mean topic shift for alerted notes (0.408) exceeded non-alerted notes (0.326). Decomposing by category overlap with a writer's five preceding notes reveals that non-alerted notes split into two modes, one near zero when writers stayed within their recent category and one near 0.45 when they explored new categories, while alerted notes showed a unimodal pattern with few near-zero values, indicating that alerts systematically pulled writers away from familiar ground (Figure 1). At the category level, writers under alerts shifted toward Politics & Conflicts (+6.03 pp) while reducing attention to Policy Violation (−3.33 pp), as shown in Figure 2.

At the aggregate level, however, these shifts concentrated contributions in already dominant categories. Politics & Conflicts grew from 38.2% to 46.6% of notes (+8.38 pp), while Health (−5.28 pp) and Policy Violation (−3.36 pp) lost coverage (Figure 3).

To assess effects on visibility, we fitted a mixed-effects model with random intercepts for contributors and fine-grained topics. We found that request alerts were associated with higher probabilities of achieving CRH status across all categories, with adjusted gains ranging from 5 pp to 15 pp, as shown in Figure 4. However, Politics & Conflicts, the category receiving the most alerts, showed only modest gains (+6 pp), suggesting consensus difficulty constrains visibility improvements even when demand is signaled.

Finally, visibility gains depended on domain familiarity (Figure 5). Without alerts, notes reflected writers’ self-selected interests, and CRH rates remained stable regardless of topic shift (slope = 0.020). With alerts, writers who stayed near their expertise saw the highest CRH rates, but visibility declined steadily as they ventured further from familiar ground (slope = −0.151).

Implications. Our findings demonstrate that request alerts are effective at the individual level: they steer contributors toward content they would not otherwise address and increase the likelihood of achieving CRH status. Yet at the system level, these same alerts concentrate collective effort in already dominant categories. This individual-aggregate paradox reveals that demand signals successfully diversify what any single contributor writes, but aggregate demand patterns narrow what the system as a whole covers. Furthermore, while request alerts increase note visibility across all topic categories, this benefit diminishes as writers move further from their prior interests, suggesting alerts are most effective when they activate existing expertise rather than drawing contributors into unfamiliar territory. Platforms seeking to improve both diversity and visibility in crowdsourced fact-checking may need to move beyond uniform alert broadcasting toward expertise-aware routing and algorithmic promotion of underserved topics.

References

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Category	Representative keywords	Note count		Publication rate	
		request-alerted	non-alerted	request-alerted	non-alerted
Politics & Conflict	Trump, voting, war, Gaza	4,862	17,017	18.94%	10.94%
Policy Violation	gambling, altered, clickbait, fake	2,260	11,113	30.49%	15.28%
Social & Civil Issues	protest, crime, gang, immigrant	1,402	5,303	22.61%	12.62%
Health	COVID, autism, vaccine, disease	426	4,152	16.90%	16.28%
Science & Technology	NASA, energy, food, chromosome	552	3,387	34.60%	17.04%
Entertainment	artist, NBA, game, Olympics	582	2,105	29.04%	17.91%
Business & Finance	stock, market, tariff, bitcoin	348	1,365	16.09%	9.60%
Total		10,432	44,442	23.15%	13.48%

Table 1: Breakdown of note counts and publication rates by category, stratified by request alert status. Publication rates show the proportions of notes that currently hold CRH status.

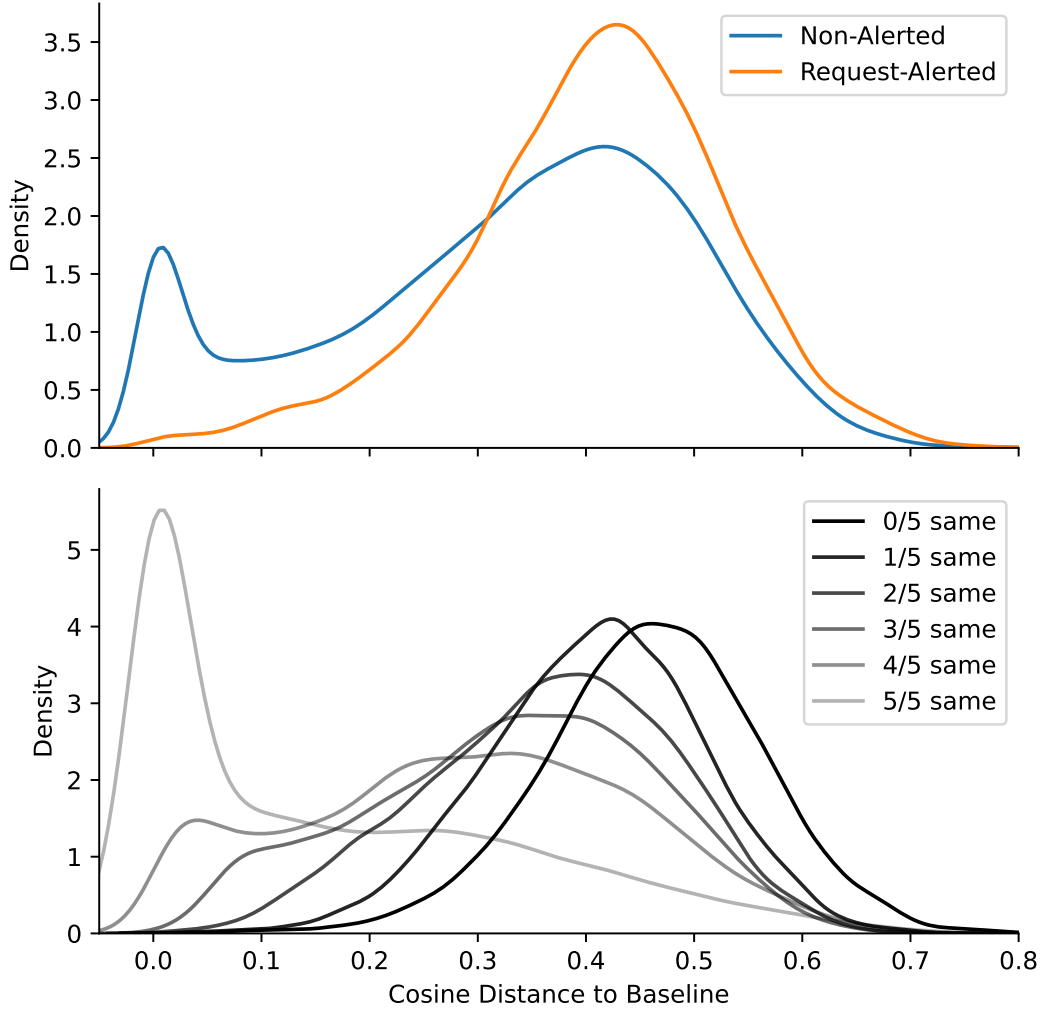


Figure 1: Comparison of topic shift distributions for notes written with and without request alerts. Top: non-alerted notes exhibit a bimodal pattern, while request-alerted notes show a unimodal distribution shifted away from zero. Bottom: decomposition by category overlap with the writer’s five preceding notes. Darker shading (0/5 same) indicates the note falls in a completely different category from all five preceding notes; lighter shading (5/5 same) indicates full category consistency. The bimodal pattern in non-alerted notes arises because writers either stay within their recent category (low shift, high overlap) or occasionally explore entirely new categories (high shift, low overlap).

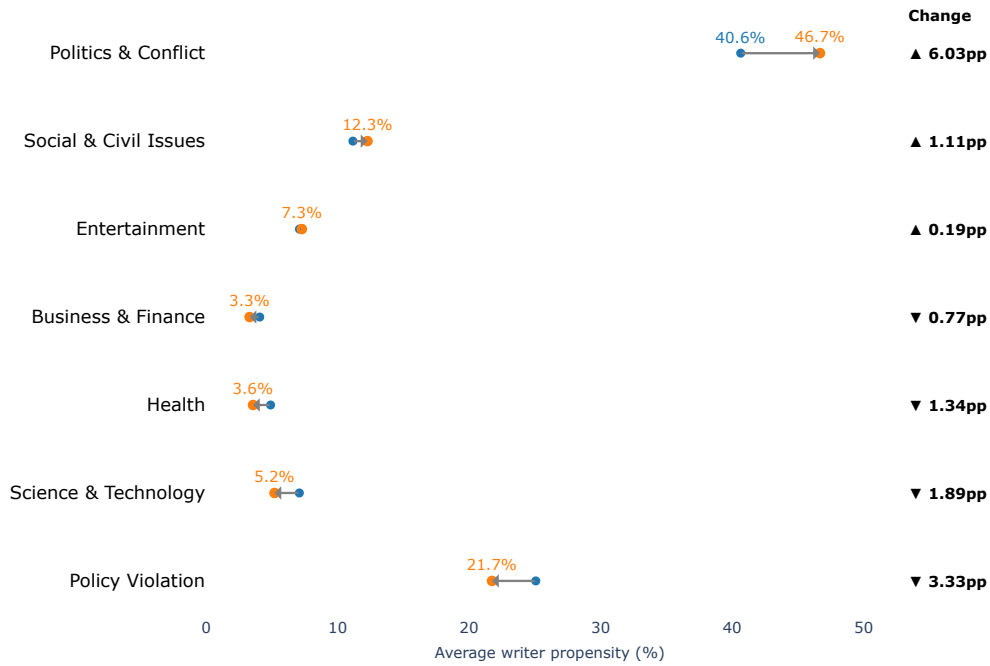


Figure 2: Within-writer topic distribution shift under request alerts. Each writer’s contribution is weighted equally: we computed the proportion of their notes in each category with and without alerts, then averaged across writers. Blue circles represent average writer propensity without alerts; orange circles represent average writer propensity with alerts. ▲ indicates categories receiving more attention under alerts; ▼ indicates less attention.

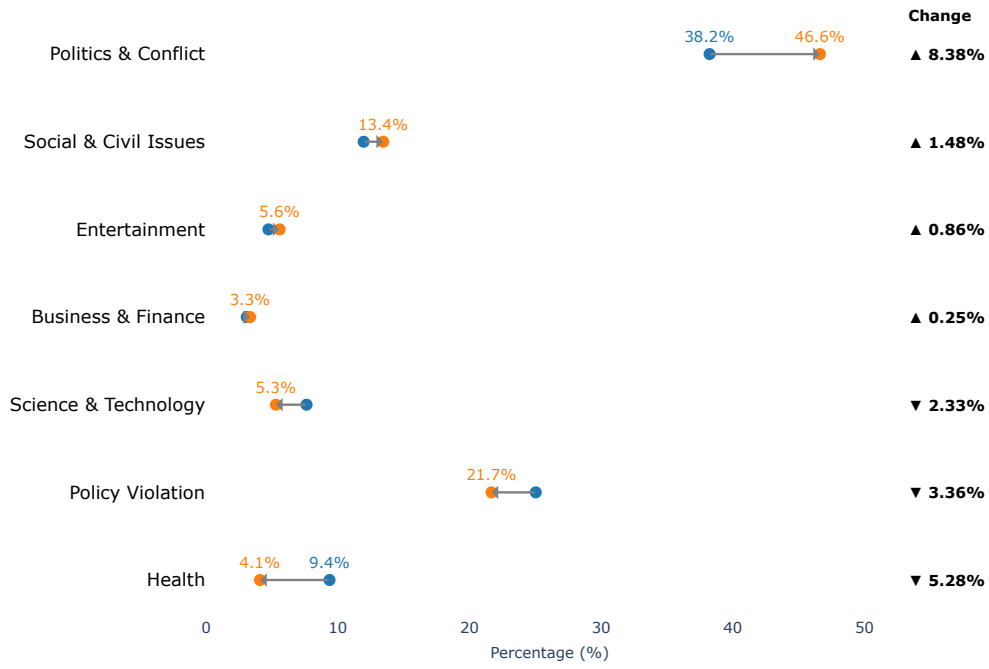


Figure 3: Aggregate note-level category distribution by request alert status. Each note is counted equally regardless of author, reflecting the combined effect of individual shifts and differential writing volume. Blue circles represent note written without alert; orange circles represent note written with alert. Request-alerted notes concentrate in Politics & Conflict (+8.38 pp), while Health (−5.28 pp) and Policy Violation (−3.36 pp) lose coverage, illustrating how demand-driven mechanisms can amplify existing topic imbalances at the system level.

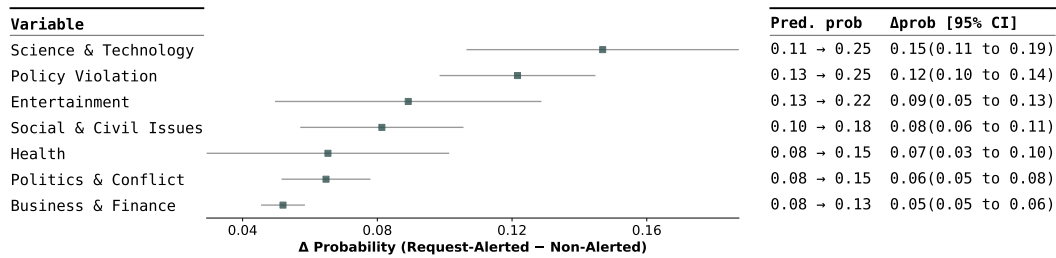


Figure 4: Effect of request alerts on note publication by note category. Predicted probabilities and 95% credible intervals estimated from Bayesian mixed-effects logistic regression with random intercepts for contributor and fine-grained topic.

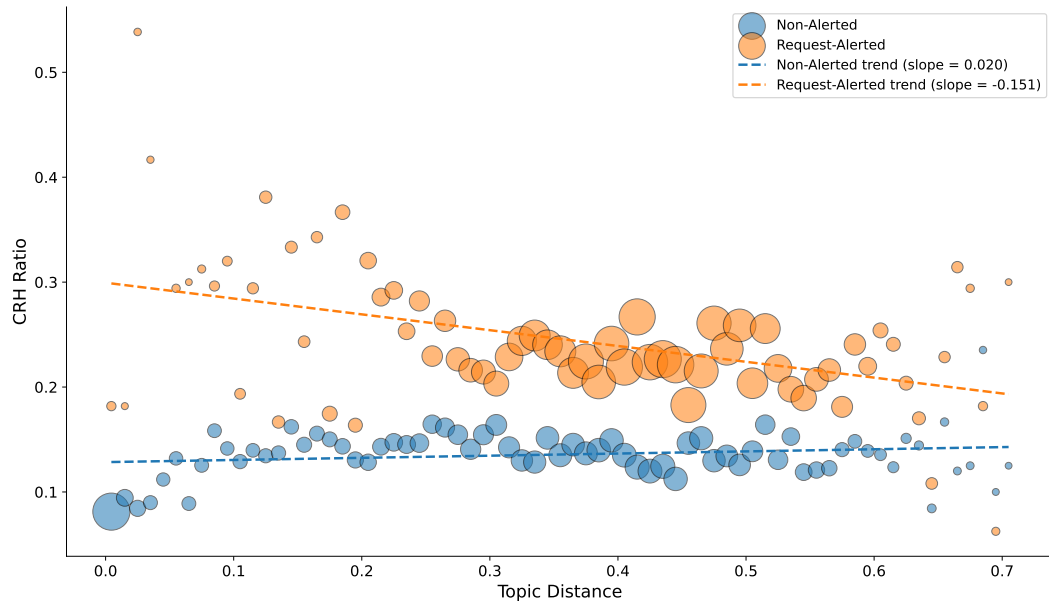


Figure 5: CRH ratio as a function of topic shift, stratified by request alert status. Non-alerted notes maintain stable CRH rates regardless of topic shift. Request-alerted notes show declining CRH rates as topic shift increases, indicating that visibility gains attenuate when alerts draw writers beyond their domain expertise.